Spectrum of the Laplacian on the Page metric

Yiqing Wang

Department of Physics and Astronomy

Undergraduate Summer Research Symposium

Eigenvalues and eigenfunctions

- If $\mathbf{L}$ is a real, symmetric matrix, we say λ is an *eigenvalue* if

$$\mathbf{L}v = \lambda v$$

  The set of L.I. eigenvectors v form a basis for the vector space.
- Let ∇^2 represent the Laplace operator. In normal $\mathbb{R}^3$ it is

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

- An eigenvalue and eigenfunction of ∇^2 satisfies

$$\nabla^2 u = -\lambda u$$

  The set of allowed λ is called the **spectrum.**

Why is the spectrum important?

If we solve the wave equation in $\mathbb{R}^3$:

$$\left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2}+\nabla^2\right)\Psi=0$$

After separating variables in spherical coordinates (R,θ,ϕ)

$$\Psi=T(t)R(r)Y(\theta,\phi)$$

we end up with an eigenvalue problem on S^2:

$$\nabla^2_{S^2}Y=-\lambda Y$$

where λ is a separation constant and $\nabla^2_{S^2}$ is the Laplacian on S^2.

Spherical harmonics

$$\frac{1}{Y}\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial Y}{\partial\theta}) + \frac{1}{Y}\frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2} = -\lambda$$

This is a well studied problem (Math 3C03!). We separate variables: $Y = P(\theta)Q(\phi)$ and use boundary conditions. The spectrum is

$$\lambda = -\ell(\ell+1), \qquad \ell = 0, 1, 2, 3, \ldots$$

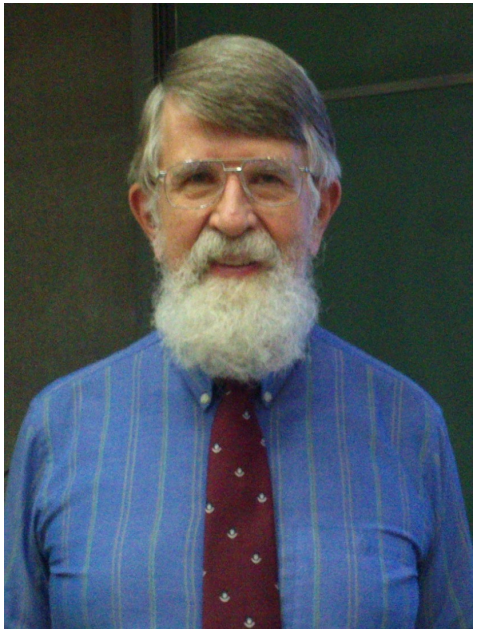
and the eigenfunctions are

$$Y_\ell^m = P_\ell^m(\cos\theta)e^{im\phi}, \qquad |m| \leq \ell$$

The spectrum gives important information about the geometry of the sphere: these are the 'vibrational frequencies'. One finds $T(t) \propto \sin\sqrt{\lambda}ct$.

My project: Page's metric

The geometry of the sphere is special (it is an example of an **Einstein manifold**) in 2D. Such spaces are very special in geometry.



- **Don Page** FRSC, an American-born Canadian theoretical physicist at the University of Alberta, Canada

The physicist Don Page found a new 4D Einstein manifold (1979) - 1st non-trivial, non-spherical example .[1] He arrived at this from studying black hole solutions of the Einstein equations. The underlying manifold is two-sphere 'twisted' over another two-sphere.

An open question is to **obtain the spectrum of the Laplace operator** on this space.

Separation of variables

The Page manifold is 4-D. Amazingly one can separate variables to reduce eigenvalue problem to a single Sturm Liouville problem!
After separation of variables , the final Sturm-Liouvilleequation is :

$$\frac{1}{S\sqrt{B(x)}}\partial_x[A(x)\sqrt{B(x)}\partial_x u(x)] - [\frac{n^2}{4\alpha^2 SA(x)} + \frac{\mu}{4SB(x)}]u(x) = -\lambda u(x)$$

where

- $u(x)$ is the eigenfunction
- λ is the eigenvalue
- $A(x)$, $B(x)$ are complicated known functions
- S,α are constants
- $\mu = \ell(\ell+2) - n^2$ where $\ell = 2k + |n|$
- $n, k = 0, 1, 2, 3, \ldots$ is similar to m, n in spherical harmonics

Numerical strategy

We can use Chebyshev or Fourier series to appoach it numerically.[2]
Let

$$u(x) = \sum_{n=1}^{\infty} c_n v_n(x)$$

Then our equation becomes :

$$\nabla^2 u(x) = -\lambda u(x) \quad \Rightarrow \quad \nabla^2 \sum_{n=1}^{\infty} c_n v_n(x) = -\lambda \sum_{n=1}^{\infty} c_n v_n(x)$$

Since the basis functions are orthogonal, it can be rewrited as:

$$\nabla^2 v_n(x) \vec{a_n} = -\lambda v_n(x) \vec{a_n}$$

Which $\vec{a_n}$ is the coodinate of corresponding eigenfunctions

Numerical strategy

Finally, discretize our domain $x = [0, \delta x, 2\delta x, \ldots, x_m ax - \delta x, x_m ax]$
Define matrices

$$\mathbf{D} = D_{ab} = \nabla^2 v_b(x_a) \qquad \mathbf{\Phi} = \Phi_{ab} = v_b(x_a)$$

then:

$$\nabla^2 v_n(x)\vec{a_n} = -\lambda v_n(x)\vec{a_n} \quad \Rightarrow \quad \mathbf{D}\vec{a_n} = \lambda \mathbf{\Phi}\vec{a_n} \quad \Rightarrow \quad \mathbf{\Phi}^{-1}\mathbf{D}\vec{a_n} = \lambda \vec{a_n}$$

$\Rightarrow$ eigenvalues of the matrix $\mathbf{\Phi}^{-1}\mathbf{D}$ are eigenvalues of our equation & the eigenvectors represent the coordinate of eigenfunctions in Hilbert space.
We first tested this method to numerically recover the spherical harmonics.

Results and comparison

The first few eigenvalues we obtain are:

n	k	rank:	1	2	3	4	5
0	0		0.0000	1.8525	5.5551	11.1094	18.5153
0	1		2.2015	4.1028	7.7934	13.3448	20.7496
1	0		2.4727	6.1867	11.7421	19.1465	28.4008
1	1		5.7389	9.5140	15.0818	22.4905	31.7467

Compare to the recently study of two pure mathematicians[3]:

N	Non-negative eigenvalues of $B^{-1}A$
1	1.8830
2	1.8830, 5.5789
3	1.8830, 5.5789, 11.134
4	1.8830, 5.5787, 11.131, 24.484
5	1.8830, 5.5787, 11.112, 18.094

Our result has better accuracy and can deal with different n and k, while the recent study can just compute the case $n, k = 0$.

Reference

[1] Page, D. (1978). A compact rotating gravitational instanton. Physics Letters. B, 79(3), 235–238. https://doi.org/10.1016/0370-2693(78)90231-9

[2] Boyd, J. P. (John P. (2001). Chebyshev and Fourier spectral methods (2nd ed., rev.). Dover Publications.

[3] Hall, S. J., Murphy, T. (2015). Erratum to: On the spectrum of the Page and the Chen–LeBrun–Weber metrics. Annals of Global Analysis and Geometry, 47(2), 223–223. https://doi.org/10.1007/s10455-015-9451-7